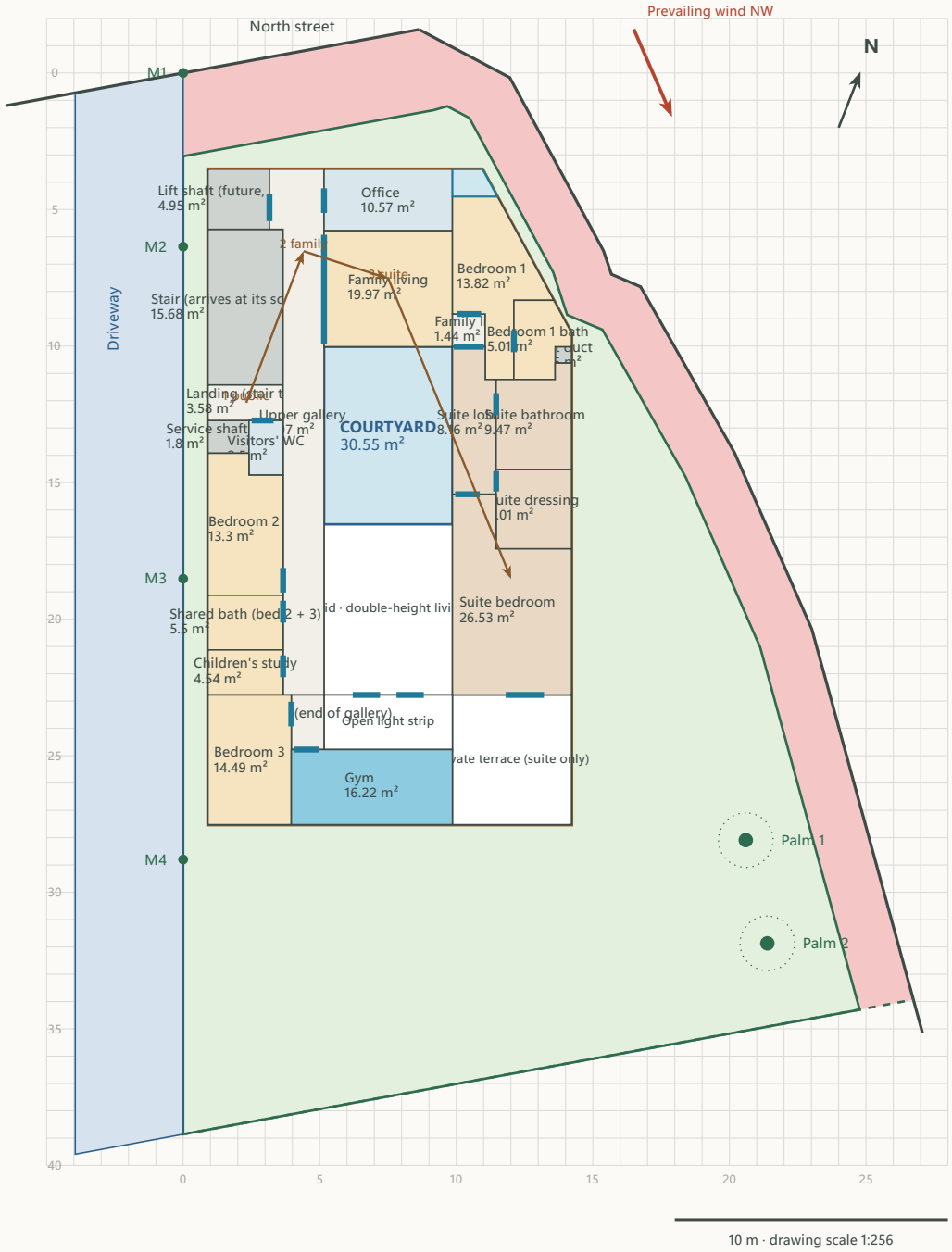


10 m · drawing scale 1:256

Taif / Palm House

First floor V4.1 · baseline · zoning and first room layout

17 Oct 2026 · First floor V4.1 on ground floor V4.0 · rooms in data/rooms_ff.csv · details to follow · furnished plan, table and light-strip check on the next pages

First floor V4.1 (baseline). The men's majlis roof is used: bedroom 3 (west), the gym (middle) behind a 2.0 m open light strip in front of the double-height living, the suite's private terrace (east). The suite lobby now opens to the bedroom, dressing and bathroom. Bedroom 1 grew round the east duct; the office became a 2.25 m strip. Colours: master suite (beige) · family (yellow) · office and visitors' WC (blue-grey) · gym (blue) · circulation (light) · fixed ground-floor elements (grey) · voids, terrace and light strip (white). Blue marks: doors and the two clerestory windows. Areas: available 225.6 m² · rooms 195.49 m² (circulation 45.55 m² + 4.23 m² route inside the family living) · terrace + light strip 30.11 m².

✓ **All checks passed**

Areas

Envelope	**667.87 m²**
Building limit (envelope unless set)	**309.68 m²**
Other open voids	**59.49 m²**
Courtyard (open to sky)	**30.55 m²**
Rooms placed (22)	**218.27 m²**
Limit minus courtyard and voids	**219.64 m²**
Air channel + duct	**1.37 m²**
Not yet assigned	**1.37 m²**

Orientation

True compass vs the M line	**21.4°**
NW wind vs the M line	**23.6°**

Concept only. Wind: Global Wind Atlas (area data, 10 m), NW ≈ 15–20 %, SE ≈ 7–8 %. Confirm on site with a ribbon test.

Schedules

All coordinates in metres from M1: x east (square to the M line), y along the M line southwards.

Envelope corners

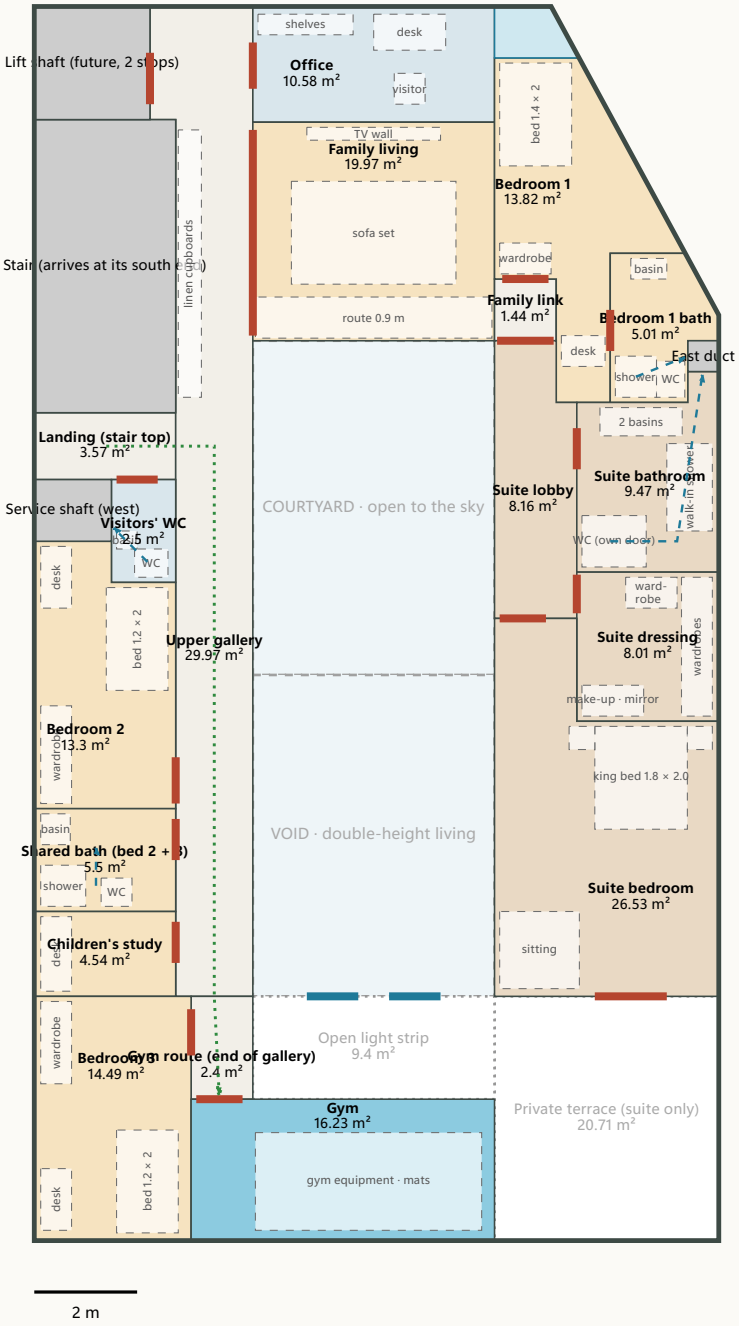
Corner	x	y
V1	0	38.86
V2	0	3.05
V3	9.18	1.36
V4	9.67	1.22
V5	10.48	1.65
V6	13.55	7.3
V7	14.06	8.86
V8	15.35	9.4
V9	18.4	14.8
V10	21.13	21.04
V11	24.77	34.3

Rooms

Name	Frame	w × d (m)	Area
Stair (arrives at its south end)	M	2.75 × 5.7	15.68 m²
Lift shaft (future, 2 stops)	M	2.25 × 2.2	4.95 m²
Service shaft (west)	M	1.5 × 1.2	1.8 m²
East duct	M	0.6 × 0.6	0.36 m²
Landing (stair top)	M	2.75 × 1.3	3.58 m²
Upper gallery	M	2 × 19.25	29.97 m²
Gym route (end of gallery)	M	1.2 × 2	2.4 m²
Visitors' WC	M	1.25 × 2	2.5 m²
Bedroom 2	M	2.75 × 5.2	13.3 m²
Shared bath (bed 2 + 3)	M	2.75 × 2	5.5 m²
Children's study	M	2.75 × 1.65	4.54 m²
Bedroom 3	M	3.05 × 4.75	14.49 m²
Gym	M	5.9 × 2.75	16.22 m²
Office	M	4.7 × 2.25	10.57 m²
Family living	M	4.7 × 4.25	19.97 m²
Family link	M	1.2 × 1.2	1.44 m²
Bedroom 1	M	3.709 × 6.7	13.82 m²
Bedroom 1 bath	M	2.111 × 2.9	5.01 m²
Suite lobby	M	1.6 × 5.4	8.16 m²
Suite bathroom	M	2.761 × 3.9	9.47 m²
Suite dressing	M	2.761 × 2.9	8.01 m²
Suite bedroom	M	4.361 × 7.35	26.53 m²

First floor V4.1 · furnished plan

CSV coordinates (x east, y south) · wall centre lines · furniture as thin outlines · red = doors · blue = clerestory windows · blue dashed = drains · green dotted = gym route · north up



Furnished test

- **Master suite:** the suite door opens into the lobby, the distributor: the bathroom (2 basins, walk-in shower, WC in its own compartment), the dressing (wardrobes for two, make-up table and full mirror) and the bedroom each open from it. The bedroom door is at the lobby's south end; the king bed (1.8 × 2.0, two bedside tables) stands against the bedroom's north wall, east of the door line, facing south to the terrace and the palms, so it is not seen from the suite door. Sitting corner by the solid west wall (the double-height void is behind it: no window there). Terrace from the bedroom only.
- **Bedroom 1** (≈ 13.82 m²) now wraps round the east duct; its bath (≈ 5.01 m²) sits on the duct's west face, the suite bath on its south face.
- **Children:** bedroom 2 and bedroom 3 with bed 1.2 × 2.0, wardrobe and desk; door swings clear of the furniture; the shared bath between them; a small children's study opens from the gallery.
- **Family living:** seating 15.75 m² (sofa set + TV wall) outside a 0.9 m route along the courtyard edge to bedroom 1 and the suite.
- **Office:** desk + one visitor seat + shelves, door from the gallery near the stair top. **Visitors' WC** from the landing (kept).
- **Gym:** reached from the end of the upper gallery by a 1.2 m route (green), past no bedroom and not the suite, beside (not across) the light strip.

Drain runs (blue dashed)

- visitors' WC: ≈ 1 m to the west shaft
- shared bath: directly over the GF family WC / serving passage, short drop into the GF west wash block run
- bedroom 1 bath: ≈ 1 m to the east duct (west face), in the floor build-up
- suite bath: ≈ 3 m in a sunken bath floor to the duct (south face), not in the kitchen ceiling (hood zone)

No drain over the master bedroom or in the ground-floor kitchen ceiling.

First floor V4.1 · rooms, circulation, distances, walls, problems

Room	Axes w × d (m)	Axes m²	≈ clear m²
Landing (stair top)	2.75 × 1.3	3.57	2.73
Upper gallery	polygon	29.97	25.67
Gym route (end of gallery)	1.2 × 2	2.4	1.76
Visitors' WC	1.25 × 2	2.5	1.85
Bedroom 2	polygon	13.3	11.58
Shared bath (bed 2 + 3)	2.75 × 2	5.5	4.5
Children's study	2.75 × 1.65	4.54	3.62
Bedroom 3	3.05 × 4.75	14.49	12.73
Gym	5.9 × 2.75	16.23	14.35
Office	4.7 × 2.25	10.58	9.07
Family living	4.7 × 4.25	19.97	18.18
Family link	1.2 × 1.2	1.44	0.96
Bedroom 1	polygon	13.82	11.78
Bedroom 1 bath	polygon	5.01	4.01
Suite lobby	polygon	8.16	6.76
Suite bathroom	polygon	9.47	8.04
Suite dressing	2.76 × 2.9	8.01	6.8
Suite bedroom	polygon	26.53	24.06
Rooms		**195.49**	
Private terrace (suite only)	4.36 × 4.75	20.71	—
Open light strip	4.7 × 2	9.4	—
Available first-floor area		**225.6**	
Used (rooms + terrace + light strip)		**225.6**	

Clear ≈ axis area minus half a wall along every edge (250 mm outer, 200 mm inner), corners ignored; open edges (landing, family living to the gallery) are counted as walls.

Family living: 19.97 m² = seating 15.75 m² + route 4.23 m² (0.9 m along the courtyard edge).
Circulation: landing + gallery + gym route + family link + suite lobby = **45.55 m²**, + 4.23 m² route inside the family living = **49.78 m²**.
Bedroom door → its bathroom door: bedroom 2 ≈ 1.3 m · bedroom 3 ≈ 4 m (via the gym route and the gallery) · bedroom 1: en-suite, door inside · master: bedroom door → lobby → bath door ≈ 3.5 m.
Men's majlis roof (x 2.75–13.311, y 19.25–24): bedroom 3 strip x 2.75–3.05 · gym route 2.4 m² · gym 16.23 m² · light strip 9.4 m² · terrace 20.71 m².

Walls off the ground-floor wall lines (for the engineer)

Ground-floor lines: x 0, 2.75, 4.25, 8.95, 13.311 · y 0, 6.5, 13.0, 19.25, 24 (and in the west band y 7.9, 10.4, 12.9, 14.4, 16.1, 17.5). Off those lines:

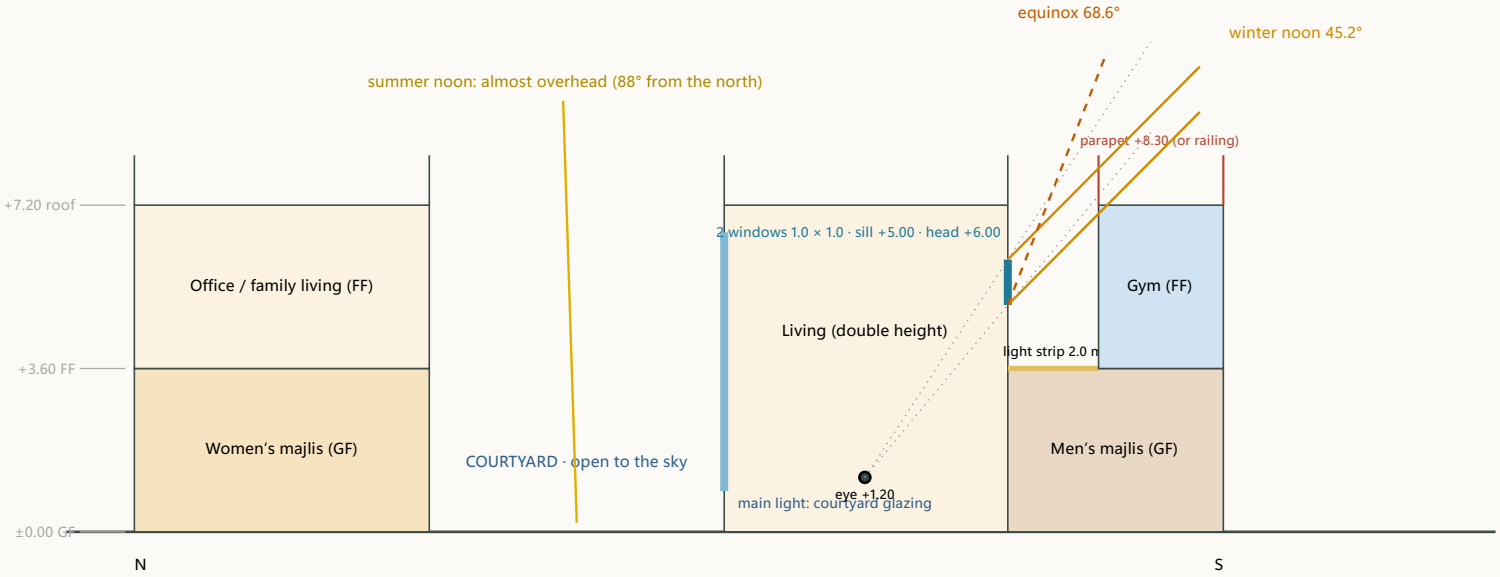
- **North band (over the women's majlis):** y 2.25 (office / family living).
- **West band:** x 1.5 and y 11.2 (visitors' WC), y 15.6 and 17.6 (shared bath).
- **Over the men's majlis** (on its roof slab, no wall below): x 3.05 (bedroom 3 / gym route), y 21.25 (gym and gym route / light strip), x 8.95 between the gym and the terrace (a parapet / screen).
- **East band:** the family link (x 10.15), bedroom 1 and its bath (x 11.2, y 4.8, y 7.7, x 12.711), the suite (x 10.15–10.55, y 7.7, 11.0, 11.9, 13.9).
- **Load and vibration check by the engineer:** the gym and bedroom 3 over the men's majlis (and the walls they need there).

Problems still open

1. **Bedroom 3 is 3.05 m wide, not 4.25 m:** the gym can only be reached from the gallery end along the 1.5 m strip of majlis roof west of the light strip; a 1.2 m route takes it. Bedroom 3 is now a normal proportion (3.05 × 4.75 = 14.5 m²). A 4.25 m bedroom 3 would mean reaching the gym through bedroom 3, or across the light strip.
2. **Family seating 15.75 m²** (target ≥ 16): the band over the women's majlis is 6.5 m deep for the office (2.25 m), the seating and the 0.9 m route.
3. **Office ≈ 9.07 m² clear**, a 2.25 m deep strip (desk + one visitor seat fit).
4. **Suite lobby** is ≈ 8.16 m² (a 1.6 m wide distributor 4.2 m long plus the 1.2 m entry: the price of three doors off it and the bent entry); the suite bedroom is ≈ 26.53 m².
5. **The gym blocks the winter sun on the clerestory** (page 5): fix options are a railing instead of a parapet on the gym roof, or a lower gym roof.
6. **Children's rooms** stay 2.75 m (bedroom 2) and 3.05 m (bedroom 3) wide; furniture fits, beds along the length.

First floor V4.1 · light-strip check · the double-height living's south windows

section north → south through the middle band (CSV x ≈ 6.6) · levels to confirm: GF ±0.00 · FF +3.60 · roof +7.20 · parapets 1.10 · a full daylight simulation comes at a later stage



Behind the 2.0 m light strip	Winter noon sun on the windows	Equinox noon sun	Sky seen through the windows from the eye
gym with a 1.10 m parapet (top +8.30)	0 %	100 %	42 %
gym with a railing only (solid top +7.20)	81 %	100 %	100 %
no gym: full roof open (parapet +4.70 at y 24)	100 %	100 %	100 %

How the percentages are defined. "Sun on the windows": the share of the window height (sill +5.00 to head +6.00) that receives direct sun at solar noon on 21 Dec (winter) or 21 Mar / 23 Sep (equinox), measured in this section through the windows; the ray from each point of the window must pass over the top of the gym's north wall 2.0 m away. "Sky seen": from a seated eye 1.20 m high in the middle of the double-height living floor (CSV y 16.1), the share of the vertical angle under which the windows are seen that shows sky rather than the gym. Pure geometry: no reflections, no sky model. A daylight simulation comes later.

- **The main light of the double-height living comes from the courtyard** (its north glazing, on the courtyard side). The south windows are small: two of 1.0 × 1.0 m (sill +5.00, head +6.00), for extra light and cross ventilation (warm air out high on the south).
- **With the gym and a 1.10 m parapet,** the 2.0 m strip still gives sky light and the equinox sun, but no winter noon sun on the windows.
- **With a railing on the gym roof** (no solid parapet), most of the window gets the winter sun again.
- **Summer:** the sun is almost overhead; the windows get sky light and light reflected from the strip, so give the strip and the gym's north wall a light colour; a small hood (≈ 0.4 m) over each window.
- **Proposal:** the 2.0 m strip, two small windows, and a railing (not a parapet) on the north edge of the gym roof. Confirm with a daylight simulation later.